MILITARY SPECIFICATION

HYPODERMIC INJECTION APPARATUS, JET, AUTOMATIC

This specification is mandatory for use in all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of hydraulically operated, hypodermic, jet injection apparatus for rapidly injecting, at high pressure and velocity through a jet tip, metered quantities of immunizing agents in such a manner that the immunizing agent (vaccine) makes its own opening into the skin and penetrates to a suitable depth for immunization.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-N-196	Needles, Hypodermic, for Luer Syringes.
QQ-A-225/6	Aluminum Alloy Bar, Rod, and Wire, Rolled, Drawn, or Cold-Finished, 2024.
QQ-A-200/3	Aluminum Alloy Bars, Rods, and Shapes, Extruded, 2024.
QQ-A-250/8	Aluminum Alloy Plate and Sheet, 5052.
QQ-A-250/4	Aluminum Alloy Plate and Sheet, 2024.
QQ-A-367	Aluminum Alloy Forgings, Heat Treated.
QQ-S-763	Steel Bars, Shapes, and Forgings-Corrosion Resisting.
QQ-S-766	Steel Plates, Sheets, and Strip, Corrosion Resisting.
QQ-W-423	Wire, Steel, Corrosion Resisting.
PPP-B-636	Box, Fiberboard.

FSC 6515

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Military

MIL-T-152	Treatment, Moisture-and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-G-23827	Grease, Aircraft and Instruments, Gear and Actuator Screw
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, and Nickel Alloys.
MIL-P-8013	Plastic Materials, Polyester Resin, Glass Fiber Base, Low Pressure Laminated.
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-M-15071	Manuals, Equipment and Systems

STANDARDS

Air Force-Navy Aeronautical

AN816	Nipple-Flared Tube and Pipe Thread.
AN6227	Packing, O-Ring Hydraulic.
AN6270	Hose Assembly-Detachable Swivel Fitting, Low Pressure.

Military

[illegible]-STD-10	Surface Roughness, Waviness and Lay.
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MS16686	Medical Services Symbol.
MS20450	Rivet, Tubular.
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure.
MS28782	Retainer, Packing, Back-Up, Teflon.

DRAWING

Defense Personnel Support Center

20241	Hypodermic Injection Apparatus, Jet, Automatic, 110 Volts, 60 Cycle, AC (9 Sheets) and Bill of Materials.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B149-52 Leaded Nickel Brass (Leaded Nickel Silver) and Leaded Nickel Bronze (Leaded Nickel Silver) Sand Castings.

(Application for copies should be addressed to the American Society for Testing and Materials,1916 Race Street, Philadelphia Pa. 19103).

AMERICAN IRON AND STEEL INSTITUTE

Steel Products Manual, Stainless and Heat-Resisting Steels.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, N.Y. 10017).

NATIONAL FIRE PROTECTION ASSOCIATION

National Electrical Code.

(NFPA Standards are available from the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts, 02110.)

UNDERWRITERS' LABORATORIES, INC.

Underwriters' Laboratories Standards.

(Underwriters' Laboratories Standards are available from the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, L.I., N.Y., 11749, 207 East Ohio Street, Chicago Ill. 60611, and 1655 Scott Boulevard, Santa Clara, Calif. 95050).

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The subject item shall comply with the fire and casualty hazards provisions of the applicable standards and be an approved type for the purpose intended, as determined by a nationally recognized testing agency, adequately equipped and competent to perform such services, as shall be evidenced by the attachment of its seal, stamp or label to the article, or issuance of a certificate. The stamp, label, seal or certificate of the Underwriters' Laboratories (New York, N.Y.), the United States Testing Laboratories, Inc. (Hoboken, New Jersey), the Electrical Testing Laboratories,

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Inc. (New York, N.Y.), or other testing laboratory approved by the contracting agency shall be considered acceptable evidence of such approval; however, in addition to the aforementioned, the contracting agency reserves the right to require the contractor to submit a laboratory test report which shall include test results and methods used to determine conformance.

3.1.2 Compliance with the above requirements regarding fire and casualty hazards, does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.2 First Article. Unless otherwise specified, prior to the start of production of the contract quantity, at least two samples of the hypodermic injection apparatus shall be assembled, completely finished. One sample shall be submitted, together with the contractor's inspection and test reports, to the procuring contracting officer for examination and testing to determine compliance with the requirements of this specification; the second sample shall be submitted for testing in compliance with the requirements of 3.1.1. Written approval of the materials, construction, and performance of the sample shall be obtained from the procuring contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Pending approval of the first articles, acquisition of materials or components, or commencement of production is at the sole risk of the contractor. Accepted sample shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the sample is not approved, the contractor shall submit a new sample for approval, or modify the existent sample so as to remove the deficiencies cited.

3.3 Material. Except as otherwise authorized by the contracting officer, material shall be as specified in the bill of materials accompanying the drawing and as specified herein. Material shall be free of all defects and imperfections that may affect the serviceability of the finished product.

3.3.1 Proprietary components. When trade-named items or other proprietary components are specified, products of other manufacturers may be substituted provided that they are equal to, and interchangeable with, the item specified, and written approval of the contracting officer has been obtained.

3.4 Construction. The hypodermic injection apparatus shall be constructed strictly in accordance with the drawing, and the requirements described in this specification. The hypodermic injection apparatus shall consist of two main subassemblies: (1) an injection 'gun' and (2) a hydraulic power source. These two components shall be interconnected by means of hydraulic hoses and shall be housed within a reinforced plastic carrying case. Complete jet injection apparatus shall pass the performance test specified in 4.4.6.

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3.4.1 Injection 'gun'. The injection gun shall consist of the component parts assembled as shown in sheet 2 of the drawing.

3.4.1.1 Stock. The aluminum-alloy gun stock (part 11) shall be accurately fabricated to meet the dimensional tolerances shown on the detail drawing. Alinement of passageways, valves, and ports shall be such that all components function smoothly, without hydraulic leakage, to cock the main spring and to fire the injection. All stepped holes shall be concentric. Surfaces on which seals ride shall be equal to, or superior to, the finish specified in the drawing (see 3.4.7.1). Observation windows, calibrated from 0 to 1.0 cubic centimeters (cc), in 0.1 cc. subdivisions, shall be provided on each side of the stock. The exterior surface of the stock shall be given a satin finish (approximately 280 grit) prior to anodizing and waxed after anodizing. Particular care shall be taken to insure that rear surface of the grip of the stock is well rounded for operator comfort. All surfaces of the stock shall be given a black anodized finish as specified in 3.4.7.2. While supporting the stock in the anodizing bath, particular care shall be taken not to grasp the stock at a finished surface. The top opening in the stock shall be sealed by means of a pin plug (part 80) after the stock has been anodized. The calibration marks and identification marking as specified in 3.4.6.1 shall be filled with a white pigment after anodizing.

3.4.1.2 Valves. The stock of the 'gun' shall be fitted with three valves, an unloading or cocking valve, a firing valve, and a check valve. Valve components shall be constructed as shown in the detail drawing. The unloading or cocking valve shall permit the free circulation of the hydraulic fluid output of the pump back to the reservoir (thus unloading pump) until the cocking trigger is depressed. With the trigger depressed, the hydraulic fluid shall be forced (via the check valve) to actuate the hydraulic piston (part 24A) and compress the main spring (part 85). The firing valve shall permit the operator to release the energy stored in the main spring by opening the pressure line to the hydraulic reservoir. The check valve shall trap the fluid holding the cocked spring in place after the unloading valve trigger is released and the pump is unloaded. The ball seats on the three valves shall be properly formed and free of defects so that there is no fluid leakage past the balls when the hydraulic pressure output of the pump is 1200 pounds per square inch (psi). There shall be no creeping of the hydraulic piston (part 24A) as seen through the observation window when the unit is given a leakage test as specified in 4.4.1.

3.4.1.3 Firing spring. The firing spring (part 85) shall supply the energy required to administer the injection. The firing spring shall be made of a chrome-vanadium steel formed to have the following properties:

Type - Compression die spring
Works in hole diameter - 1 inch
Works over rod diameter - 1/2 inch

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Free length - 2-1/2 inches
Total deflection-30 percent free length
Maximum load - 600 pounds
Load per 1/8 inch deflection - 85 pounds
Ends - Closed and ground

Spring shall be the Raymond Manufacturing Company Die Spring, Catalog No. 1" X 2-1/2" H.D., or equal. The firing spring shall work over the spring guide portion of the hydraulic piston (part 24A), which shall be constructed from corrosion-resisting steel as shown in the drawing.

3.4.1.4 Hydraulic actuating device. The hydraulic piston (part 24A), seal (part 92), seal (part 91), and cylinder portion of the stock shall function to provide a hydraulic actuating device to compress the main spring. Fabricated parts shall be constructed as shown in the drawing while purchased parts shall be as described in the bill of materials accompanying the drawing. The outside diameter of the quad ring seals, parts 90, 91 and 92, shall be identified, by such means as color coding, to simplify their proper installation. If color coding is used, the coding material shall be nontoxic. Seals shall function so as to prevent leakage of hydraulic fluid from the gun stock observation windows or through the rear of the dosage adjusting screw while the injector is in use (see 4.4.6). A thin film of oil due to normal lubrication of cylinder walls as the hydraulic piston moves is normal and is not cause for rejection.

3.4.1.5 Dosage adjustment subassembly. Dosage adjustment from 0.1 to 1.0 cc. shall be accomplished by means of the dosage adjustment subassembly comprising parts 26, 27, 28, 38, 81, 82, and 99. The threads of the adjusting stem (part 27) shall be coated with Molykote, type G. The thrust bearing (part 81), as well as the main spring (part 85), shall be coated with grease conforming to MIL-G-23827 prior to securing of this subassembly to the injection gun stock.

3.4.1.6 Vaccine pump section. The vaccine pump section shall be a self-priming, single-action pump consisting of the following component parts: vaccine piston (part 23), inlet valve housing (part 29), inlet valve (part 30), outlet valve retainer (part 31), outlet valve (part 32), cylinder (part 33), coupling nut (part 34), inlet tube (part 35), inlet valve mounting pin (part 36), key pin (part 79), valve balls (part 88), outlet valve spring (part 40), and seals (parts 90 (see 3.4.1.4), 93, 94, 96, 97). Fabricated parts shall be constructed as shown in the drawing, while purchased parts shall be as described in the bill of materials accompanying the drawing. The quad ring seal, part 90, shall be fabricated of material suitable for repeated pressure steam sterilization at 250° F. Cylinder (part 33), shall be concentric with the bore of the stock to insure accurate alignment of the injection piston (part 23) with the hydraulic piston (part 24A). Surfaces on which seals ride shall be equal to, or superior to, the finish specified in the drawing. Keying of the vaccine pump section to the gun stock shall be accomplished by a pin (part 79) in such a manner that the inlet valve assembly is in vertical

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alinement with the handle of the gun stock. The ball seats of the pump inlet and outlet valves shall be properly formed and free of defects so that positive seals are obtained during normal operation of the injector. Inlet and outlet valves shall not leak when unit is tested as specified in 4.4.2.1 and 4.4.2.2. Tube (part 35) shall be securely brazed at both ends in an airtight, waterright manner. There shall be no trace of air bubbles leaking past the brazed joint, when the unit is tested as specified in 4.4.2.1.

3.4.1.6.1 Jet nozzle. The jet nozzle shall consist of corrosion-resisting steel blank (part 41), a sapphire jewel orifice (part 42). The sapphire jewel orifice shall conform to the dimensions shown in the drawing when the jewel contour is checked by contour projection at 100 magnifications as described in 4.4.4. The mounting blank shall be fabricated in accordance with the detail drawing. The front face of the blank shall have a minimum thickness of 0.072 inch. Inside face of the blank shall have a surface equal to, or superior to, the finish specified in the drawing. The mounting hole for the sapphire jewel shall be concentric with the inside diameter of the jewel mount. The mounting of the sapphire jewel shall be by means of a friction setting, in which the bore of the housing is a minimum of 0.0002 inch smaller than the outside diameter of the jewel, followed by spinning of the blank material over the chamfer of the jewel edge. The orifice of the sapphire shall be concentric with the mounting hole in the blank. The orifice of the jewel shall be perfectly clean, free of lapping or polishing materials. The orifice of the jewel shall be regular and circular in contour and shall be free of chips or other imperfections. The face of the jewel shall be flush with the face of the blank within the dimensions specified in the drawing. A fine knurled finish shall be incorporated on the front face of the nozzle blank occupying the annular surface surrounding the mound which receives the sapphire jewel orifice. The knurl shall be of depth to provide sufficiently sharp diamond shaped projections which will grip onto skin and minimize relative slippage between nozzle and skin. The jet nozzle shall be capable of being repeatedly autoclaved in 250° Fahrenheit (F.) steam.

3.4.1.6.1.1 Protective cap. Each jet nozzle shall be provided with a drawn aluminum-alloy protective cap. The cap shall be finished in a black anodized finish as specified in 3.4.7.2. Cap shall be formed from 0.019 inch thick aluminum-alloy stock and shall conform in dimensions to the round nozzle, Hudson Tool & Die Company, Inc., Catalog No. HU-48, or equal, modified to an overall length of 5/8 inch.

3.4.1.6.2 Feed needle assembly. The feed needle assembly shall consist of the following component parts: vaccine feed tube (part 44), vent tube (part 45), vaccine tube plug (part 46), Luer hubs (parts 83, 84), filter screen (part 84A), and nylon adapter (part 73). Assembly of concentric tubes and of tubes to hubs shall be by means of sound silver-brazed joints, employing the brazing material specified in the drawing. The brazed joints shall be airtight when feed needle is tested in accordance with the procedure given in 4.4.3. Particular care shall be taken during the brazing operation not

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to distort the Luer hub (part 84) or to destroy the internal finish of this hub. Construction shall be such that an airtight joint can be made between the Luer hub (part 84) and the matching taper on the inlet valve (part 30). The filter screen (part 84A) shall be a 100-mesh monel wire screen approximately 3/16 inch diameter, suitable domed to fit securely into the vaccine inlet hub of each feed needle assembly.

3.4.1.7 Bottle holder assembly. The bottle holder assembly shall consist of the following component parts: stud (part 39), positioner (part 50), clip (part 51), jaws (part 52), jaw reinforcements (part 55), latch (part 53), and spring (part 54). Parts shall be fabricated in accordance with the details shown in the drawing. Sheet metal parts may be fabricated either of the alloys as shown in the bill of material or of type AISI 430 stainless steel. The bottle holder shall be coated with a vinyl plastisol coating material as specified in 3.4.7.3. Coating shall be applied prior to attachment of parts 39, 53, and 54. The inside surface of part 50 shall be uncoated. The jaws shall securely grasp all sizes of vaccine bottles from 10 to 60 cc. capacity. The finger-operated latch (part 53) shall operate smoothly, without binding, on the stud (part 39). The coated surface of part 51 which contacts the gun stock (part 11) shall be smoothly finished so that sliding of the bottle holder assembly back and forth on the gun stock shall not mar the black anodized finish of the stock. All exposed stainless steel parts shall be ground to a No. 4 uniform satin finish.

3.4.1.8 Multipurpose tool. Each injection unit shall be furnished with a corrosion-resistant steel multipurpose tool (part 48) constructed as shown in the drawing. The pin end of the tool shall be capable of removing and inserting valves (parts 12, 15 and 17) into the gun stock without burring, or otherwise damaging, these valves. The tool shall be capable of being slipped into the dosage adjustment assembly so that the tool handle fits into the slot of the adjustment knob (part 28) to afford additional leverage to the operator in making dosage adjustments. One end of the tool handle shall have brazed into place a needle hub wrench, constructed as shown in the drawing, capable of being used to install hub (part 84) of the feed needle assembly on the injector inlet valve (part 30) in a leakproof connection.

3.4.2 Hydraulic system. The hydraulic system shall consist of a hydraulic power unit, hydraulic fittings, and hoses.

3.4.2.1 Hydraulic power unit. The hydraulic power unit shall consist of a single compact unit comprising an electric motor with starting relay and capacitor, a gear-type pump, pump to motor coupling, an integral reservoir with filter, a relief valve, and a pressure switch. The power unit shall be the Oil-Dyne, Inc. model NTOT-6. The power unit shall be dimensioned so that it will fit within the hydraulic unit housing (parts 1 and 2) to yield an assembly as shown on sheet 1 of the drawing. The maximum weight of the complete power unit shall be 10 pounds. The power unit components and the assembly shall pass the acceptance test specified in 4.4.5.

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3.4.2.1.1 Electric motor. The electric motor shall be General Electric motor, Model 5 KC15 FG; a 1/12 horsepower, 3450/2850 revolutions per minute, 115-volt, 60/50-cycle, single-phase, open-frame construction, all-angle sleeve bearing, no base, heavy nose and shield, 5/16 inch diameter by 1-3/8 inches long plain round shaft, counterclockwise rotation (looking from motor toward shaft extension). There shall be three 12 ± 1 inch leads out of the top of the shell opposite the shaft end. The motor leads shall be braidless neoprene leads, color coded as shown on the wiring diagram, sheet 1 of the drawing, rated 90° Centigrade (C) in air. The motor shall have a continuous temperature rating of 40° C., shall be antifungus treated as specified in MIL-T-152, and shall be finished to meet the requirements of MIL-E-5272. The motor dimensions shall be equal to those of General Electric Motor outline 111A686-003, except for the shaft modification to accept the pump coupling. The motor-starting relay shall be of the current type, Essex Wire Corp., REM Division, Part No. PN91252-313. The starting relay shall be treated to be fungus resistant. The capacitor shall be the P. R. Mallory & Co., Inc.'s, type P, Motor Start Capacitor, 108 microfarads, at 110 volts, alternating current, 1-7/16 by 2-3/4 inches, with two 6-inch type B plastic-covered leads, complete with PLA-3 end caps and HB-2 bracket.

3.4.2.1.2 Hydraulic pump. The hydraulic pump shall be of the high-pressure, gear type, capable of continuous duty at 3000 psi and up to 5000 psi peak. The pump shall deliver 17 ± 2 cubic inches of hydraulic fluid per minute at 1200 psi when the motor is being operated from 60-cycle, 115-volt, alternating current lines. The pump shall be coupled to the motor with a leakproof seal between the motor and pump, thus permitting the unit to be operated or shipped in any position, with oil in the reservoir, without danger of oil leaking into the motor. The hydraulic pump outlets shall come off the sides of the unit and shall be 1/8-inch female pipe threads. Surfaces of outlet ports shall be suitable for proper sealing of Tru-Seal fittings (see 3.4.2.1.7).

3.4.2.1.3 Hydraulic reservoir. The hydraulic reservoir shall act as an enclosure for the hydraulic pump. The fluid capacity of the reservoir shall be approximately 40 cubic inches. Within the reservoir, on the pump inlet line, there shall be a filter, 1/2 gallon per minute capacity, capable of filtering all fluid entering the pump to 10 microns or less. The filter shall be ultrasonically cleaned before installation. The reservoir shall be provided with a breather which shall permit the power unit to be held and operated in any position without fluid leaking from the breather. An adequate-sized filling plug shall be provided which shall make possible the filling of the reservoir with hydraulic fluid while the power unit is mounted in the hydraulic unit housing (part 2) as shown on sheet 1 of the drawing.

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3.4.2.1.4 Relief valve. The relief valve shall be mounted externally and shall be capable of being adjusted with the power unit bolted to the hydraulic unit housing (part 2). The relief valve shall be fitted with a spring capable of being adjusted so that the relief valve operates in the 1000 to 2000 psi range. Relief valve shall be set to kick-out at 1300 ± 50 psi. Below cracking pressure there shall be zero leakage. The relief valve differential shall be ± 8 percent.

3.4.2.1.5 Pressure switch. The pressure switch shall be mounted as an integral part of the hydraulic power unit without the use of external fittings. The pressure switch shall have an electrical rating of 5 amperes, 110 volts, alternating current. The contacts shall be single-pole, normally open. A normally closed lead connection shall not be furnished. Normally open electric wire leads, type SJT, shall be supplied and shall be a minimum of 14 inches long. The pressure switch shall be in a totally enclosed closure. The pressure switch shall be fitted with a spring capable of being externally adjusted (while the power unit is bolted to the hydraulic unit housing), so that the pressure switch operates in the 1000 to 2000 psi range. The pressure switch shall be set so that it actuates approximately 50 psi below the relief valve setting of the particular unit in which it is a component. The pressure switch differential shall be ± 8 percent (see 4.4.6).

3.4.2.1.6 Hydraulic fluid. The hydraulic fluid shall be Oil Dyne O-D 18 hydraulic fluid, or equal, and shall be capable of being used in ambients ranging from -30° F. to 120° F. The storage temperature range shall be -60° F. to 150° F. Operating altitude condition shall be from -500 feet to 10,000 feet referred to sea level.

3.4.2.1.7 Hydraulic connections and hoses. Connection of the injection gun to the hydraulic power unit shall be by means of 45° elbows (part 49A), hydraulic hoses (parts 64 & 65) and nipples (part 68). Connections to the power unit shall be as shown on sheet 1 of the drawing while connections to the stock shall be as shown on sheet 2 of the drawing. Elbows and nipples shall be installed in a leakproof connection and no compound shall be used on the pipe threads. Before being connected to the jet injection apparatus, hydraulic hose assemblies shall be flushed with hydraulic fluid by connecting to a hydraulic pump with an output capacity of at least equal to the capacity of the hydraulic pump specified herein. Flushing shall continue until no foreign matter is present in the flushed fluid. Hoses shall be sealed after assembly and resealed after flushing to minimize possible entry of foreign matter into the hydraulic system.

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3.4.3 Electrical components. The electrical components, in addition to the motor (part 57B), starting relay (part 57D), and pressure switch (part 57C) and capacitor (part 57G) previously described, shall consist of a circuit breaker switch (part 59), counter (part 76), terminal block (part 114), power cord (part 60), with plug (part 75), hook-up wire (parts 66 and 67) and insulated terminal lugs (parts 61, 61A and 61B). Electrical components shall be wired as shown in the wiring diagram on sheet 1 of the drawing. The pressure switch shall actuate the counter at the instant when the firing spring is fully cocked, just prior to the relief valve operating on the hydraulic unit. The counter shall record the number of times the gun is cocked (hence the number of injections) and the counter click shall serve as an audible signal to the operator that an injection is ready to be administered. The cable clip (part 63) shall act as a strain relief for the power cable. All electric wiring and components including the pressure switch shall meet the requirements of the National Electrical Code, and complete assembly shall meet the requirement of Underwriters' Laboratories, Inc., for approved listing (see 3.1.1).

3.4.4 Sheet metal components. Sheet metal components shall be fabricated of aluminum alloy or of corrosion-resisting steel, as applicable, and as specified in the details shown in the drawing. The hydraulic unit housing (part 2), the housing top (part 1), the injection gun platform (part 5), the circuit breaker guard (part 4), the relay guard (part 43), and the electrical unit support (part 6), shall be fabricated of aluminum alloy. The hose retainer (part 3) and the clips (parts 7 and 8) shall be fabricated of corrosion-resisting steel. The housing top (part 1) and the gun platform (part 5) shall be given a black anodized finish as specified in 3.4.7.2. A layer of 1/4-inch-thick monocellular plastic (part 77) shall be cemented to the hydraulic unit housing (part 2) with pliobond adhesive (part 122), as shown in the drawing.

3.4.5 Carrying case. The reinforced plastic carrying case (part 78) shall be molded from polyester resin, glass fiber base material laminated as specified in MIL-P-8013, type I, color black. The case shall have the following inside dimensions:

	Inches
Length - - - - - - - - - - - - - - - -	13-13/16
Width - - - - - - - - - - - - - - - -	8-3/4
Height, case body- - - - - - - - - -	7
Height, case top - - - - - - - - - -	2-3/8

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A 1/2-degree taper for the case in molding is permitted. The case body and the case top shall be fitted with mating aluminum-alloy extruded valance. Aluminum shall be clear anodized as specified in MIL-A-8625 for type II. The case top shall be hinged to the body by means of a sturdy anodized aluminum-alloy continuous hinge riveted in place through the case valance. The case lid and the base of the case body shall be ribbed for strength. The case shall be fitted with a sturdy stay hinge on the right side only, with two sturdy drawbolts and with a sturdy plastic handle assembly located on the case top. On the inside of the case top there shall be provided the following components:

(a) Clip (part 62) for multipurpose tool (part 48) to be secured by one of the handle securing screws.

(b) Aluminum-alloy (black anodized) spare injection nozzle mount (part 47) secured by the other handle securing screw.

(c) A repair parts bag (part 56) securely cemented to the case as shown on sheet 1 of the drawing.

The case shall be the carrying case, series 210-1, black, H. Koch Sons, or equal, modified as shown on sheet 1 of the drawing.

3.4.6 Identification marking.

3.4.6.1 Stock. Legibly engraved on the handle of the gun stock, in the area designated on sheet 1 of the drawing (coordinate D-4), there shall appear the following information:

Hypodermic Injection Apparatus, Jet, Automatic
Serial No. _____
Manufacturer's name or registered trademark.

The serial numbers shall run consecutively for each manufacturer. The lettering shall be approximately 3/32 inch high and the engraving shall be filled with a white lacquer filling after the stock is anodized. In addition, the stock shall have engraved on each side, near the observation windows, the dosage calibration marks as shown on sheet 4 of the drawing.

3.4.6.2 Carrying case. The carrying case shall be fitted with two anodized aluminum-alloy nameplates having a thickness of 0.020 inch. One nameplate (part 107) shall measure approximately 2 by 3/4 inches and shall bear the following legend:

"Caution. For Use With 110 Volt, 60 Cycle, AC., Only."

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This nameplate shall be located on the inside of the lid of the case as described in Note 5 of sheet 1 of the drawing. Another nameplate, (part 110), shall measure approximately 4 by 2 inches and shall bear the following information:

"Hypodermic Injection Apparatus, Jet, Automatic
FSN 6515-656-1021
Manufacturer's name or registered trademark
The medical services symbol, as shown for type III, Standard MS16686"

The third nameplate, (part 125), located on the exterior rear of the carrying case, shall include the manufacturer's name, model number, unit serial number, voltage frequency rating, wattage rating, and Underwriters Laboratories approval. In lieu of this nameplate, the information may be added to nameplate (part 110). Letters on all nameplates shall be Gothic capitals of appropriate size. Characters shall be integrated into the plate by a photographic process. After the photographic processing, the image shall be sealed in the oxide layer by a suitable chemical treatment of the layer. Nameplates shall be fastened to the case by means of Pliobond adhesive. The largest nameplate shall be secured to the outside of the case, centered between the case drawbolts, with the top edge of the nameplate butting against the aluminum valance of the case body. The case surface shall be properly prepared to insure adherence of the nameplates.

3.4.6.3 Repair parts bag. The repair parts bag (part 56) shall have stencilled or printed on it, with a white waterproof paint or ink, the legend: "Repair Parts". Letters shall be spaced as shown in the detail on sheet 9 of the drawing. Repair parts bag shall be fastened to the inside of the cover with adhesive as shown on the drawing and bill of material.

3.4.7 Finish.

3.4.7.1 Machine finish symbols. Machine finish symbols shall be as defined in MIL-STD-10. Machine stainless steel parts shall be ultrasonically cleaned and passivated as specified in 3.11. Ultrasonic cleaning shall not leave a detergent film.

3.4.7.2 Black anodizing. Aluminum-alloy components which are specified to have a black anodized finish shall be finished in a heavy, uniform, continuous, black, anodic film, free of breaks, scratches, and other damage. Surfaces which are visible; i.e., the exterior of the gun stock, the top surface of the hydraulic housing top, the gun platform exterior surface, and the exterior of the protective caps shall be given a grained satin finish (approximately 280 grit), prior to anodizing. Immediately after anodizing, and before handling, anodic coated parts shall be waxed to protect the finish. Anodic coating shall comply with MIL-A-8625, type II, black-dyed coating.

3.4.7.3 Vinyl plastisol coating. Specified surfaces of the bottle holder assembly shall be coated with a black or grey vinyl plastisol coating. The coating shall be accomplished through the use of Unit Chromium Division, Metal & Thermit Corp., Unichrome Coating 5300 (Black or Grey), applied on a coat of

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Unichrome Primer 219-PX. The coating shall be applied following the manufacturer's recommended coating procedure to yield a final baked film having a minimum thickness of 5-10 mils. Finished coat shall be a tough, resilient coating free of pin-holes and blisters.

3.4.7.4 Triggers. The cast triggers (parts 20 and 21) shall be polished to round all edges and corners and then given a heavy chromium over nickel plating and polished to a satin finish.

3.4.7.5 Stainless steel parts. All exterior exposed stainless steel parts shall be given a No. 4 or better uniform satin finish. This includes part Nos. 3, 7, 8, 41, 33, 35, 29, 30, 34, 26, and 28. Part 7 used to hold the spare main spring need not be given a satin finish.

3.4.8 Repair parts. With each unit there shall be furnished the repair parts listed on sheets 16 and 17 of the bill of materials accompanying the drawing. All repair parts, with the exception of the spare jet nozzle and protective cap, main opening, and wrench, shall be placed within the repair parts bag. The repair parts bag shall be fabricated and mounted in accordance with the detail drawing covering part 56. The spare jet nozzle, with protective cap, shall be screwed onto the mount (part 47) on the inside of the case lid. Prior to being placed into the repair parts bag, the repair parts shall be segregated into groups as follows:

(1) Seals and seal back-up ring.

(2) Balls.

(3) Springs (springs shall be mounted on a corrosion-resisting steel wire loop).

(4) Spare feed needles and adapters, sharp end of feed needles shall be protected by tape or other means).

(5) Nozzle cleaning wire (part 116) and polishing cloth (part 117).

Each group shall be heat sealed in a clear plastic envelope. All sealed envelopes shall then be loaded into the repair parts bag with the high mortality parts list (part 119).

3.4.9 Weight. The finished weight of the jet injector, within the carrying case, including repair parts and manuals, shall not exceed 26 pounds.

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3.4.10 Assembly. The hydraulic power unit contained in the sheet-metal housing shall be slipped into the carrying case, care being taken to clear the valance of the case. The electric power cord shall be wrapped around the cover of the hydraulic unit housing (part 1) and then the hoses shall be similarly wrapped into place, tucked under the hose retainer (part 3). The injection gun shall be clipped onto the platform (part 5) with identification marking shown. The final assembly shall be as pictured in a photograph included in the operating instruction manual. Hoses shall be oriented following a procedure given in the manual so that they can be wrapped into the case without kinks or twists.

3.5 Tooling. The contractor shall design and manufacture or provide any tooling needed for the production and testing of hypodermic injection apparatus as required to meet specification requirements described herein. Such tooling shall remain the property of the contractor.

3.6 The contractor shall perform all services, as specified, and will not produce any of said hypodermic injection apparatus, other than in strict accordance with requirements contained in Section 3 of this specification without the previous written approval of the contracting officer.

3.7 Supplier responsibility for maintenance. The supplier shall, for a period of at least 2 years after completion of the contract, supply under contract or purchase order, such repair parts as are peculiar to his manufacture, or other commercial parts modified by him to provide components for the hypodermic injection apparatus supplied by him.

3.8 Bill of materials. There shall be provided a final Government bill of materials covering the hypodermic injection device as required by the schedule.

3.9 Interchangeability. The hypodermic injection apparatus shall be manufactured in such a manner that all removable component parts shall be interchangeable with each other and any apparatus produced under this specification.

3.10 Tests. The hypodermic injection apparatus shall perform in accordance with the following requirements.

3.10.1 Clinical tests. The Government will arrange for the clinical testing of the hypodermic injection apparatus on patients using one of the preproduction samples in order to insure the proper functioning of the sample, including injecting with a minimum of whealing or bleeding and other characteristics described herein.

3.10.2 Leakage test, hydraulic, gun. The hydraulic system shall pass the leakage test described in 4.4.1.

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3.10.3 Leakage test, air. The vaccine pump section shall meet the vaccine pump section test described in 4.4.2.

3.10.4 Leakage test, feed needle. The feed needle assembly shall meet the air leakage test described in 4.4.3.

3.10.5 Jet nozzle test. The sapphire jewel orifice shall comply with the dimensional requirements of the drawing, and, on firing, there shall be no leakage or side spray per test described in 4.4.4.

3.10.6 Hydraulic power unit test. Hydraulic power units shall comply with the tests described in 4.4.5.

3.10.7 Performance test, hypodermic injection apparatus. The injection apparatus shall fire as described in 4.4.6. In addition, the injection apparatus shall be capable of sustained continuous performance as described in 4.4.6 without failure of any sort.

3.10.8 Dielectric test. Hypodermic injection device shall pass the dielectric test described in 4.4.7.

3.11 Cleaning. Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice. Cleaning processes shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After machining, all stainless steel parts shall be ultrasonically cleaned and passivated. After assembly, components shall be cleaned thoroughly and shall be free from particles of filler metal, flux, and other foreign material. If necessary, such cleaning shall also be performed after final assembly of components. Prior to packaging, the plastic carrying case shall be rubbed down with a wax-laden cloth.

3.12 Government-furnished material. No Government-furnished material will be provided; however, the following items will be made available to the Government inspector for use in inspecting subject item:

(a) One set of gages for checking the most critical dimensions of the gun stock (part 11).

(b) One set of adapters for: (1) the inlet valve test (4.4.2.1), (2) The outlet valve test (4.4.2.2), and (3) the feed needle test (4.4.3).

At the conclusion of the contract, the contractor shall carefully check the gages and adapters, recondition them as necessary, and repack them for transport in their reusable carrying cases. Instructions for the use of the gages are included in the carrying cases.

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3.13 Technical manual. The contractor shall prepare and furnish 15 technical manuals, plus two for each equipment set (packed in a manila or white envelope placed in the carrying case). The technical manuals shall be type I, in accordance with MIL-M-15071, except approval shall be by the Defense Personnel Support Center. It should be noted that manuals shall be completed in time for Defense Personnel Support Center review and approval, and are to accompany each shipment. Manuals not accompanying equipment shall be forwarded to the contracting officer prior to completion of the contract. A list of all component repair parts subject to wear and replacement, identified by their manufacturer's name, address, model or catalog number, and current prices shall be included. An apron sheet (fold-out) of the jet gun assembly shall be provided at the rear of the manual for ready reference during review of the manual. The apron sheet shall be Kimberly Clark's No. 115, Textopoint paper, or equal.

3.14 Workmanship. Workmanship shall be first class throughout. Hypodermic injection apparatus shall be free of defects which detract from their appearance or may impair their serviceability.

3.14.1 Joining. Brazing shall be in accordance with MIL-B-7883. Brazed joints shall be durable and sound in every detail. Component parts shall be carefully and properly alined prior to brazing into permanent position. Brazed joints shall be polished smooth to blend with adjacent surfaces. Spot welds shall be free of surface burning, cracks, and porosity, and shall be tightly fused.

3.14.2 Machine work. All component parts shall be free of burs and other imperfections. All threads on threaded members shall be properly formed and of the fit specified in the drawing. Ball sets shall be properly formed and burnished to assure leaktight sealing of the balls. Alinement of components shall be such that assembly and disassembly shall be easily accomplished and such that the function of the injector is not impaired, due to uneven wearing or scratching of components caused by misalinement.

3.14.3 Hoses. Hydraulic hoses shall be clear of any rubber of foreign particles before being assembled in the injection apparatus by purging with oil (3.4.2.1.7). Woven covering shall be free of fraying. Hoses shall be attached so that they may be neatly coiled when the injection gun is mounted within the carrying case.

3.14.4 Wiring. Wiring shall be neat and secure. Wires shall be as short as practical, except that sufficient slack shall be provided to prevent undue stress or connections, to enable parts to be removed and replaced during servicing, and to facilitate field repair of broken or cut wires. Wires and cables shall be properly supported and secured to prevent undue stress on conductors and terminals and undue change in position of the wires and cables after the equipment has been subjected to specified service conditions or has been serviced or repaired in a normal manner. Clearance between connections, or conductors and metal parts, shall be at least 1/4 inch, unless use of specified parts makes such clearance impractical.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements. Prior to the submission of each lot of the injection apparatus for acceptance by the Government, the contractor will conduct and document a thorough physical examination of each unit, and make and record operational tests described herein to demonstrate conclusively the proper functioning of each unit. This inspection shall include all examinations and tests described herein as well as any other tests found necessary during the production of the units, including but not limited to the following criteria:

(a) Acceptance of preproduction sample.

(b) Compliance with Underwriters' Laboratories' requirements.

(c) Approval of materials and components.

(d) Visual and mechanical examination.

(e) Compliance with drawing and bill of materials.

(f) Approval of clinical test.

(g) Performance test.

(h) Dielectric test.

(i) Other tests described in 3.9.

Subsequent to the contractor's inspection as indicated above, an examination of the units by the authorized Government representative shall be made. Upon failure of any unit to perform, or of any component, the contractor shall replace the faulty unit or faulty component with new material. Upon completion of the testing by the Government representative as indicated above, the contractor shall carefully examine all units to insure proper condition, cleanliness, and preparation for shipment. Packaging and packing shall be included as part of this inspection.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

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4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling. Sampling for material or components shall be conducted by the Government in accordance with MIL-STD-105, as appropriate. Testing and examination of finished injection apparatus shall be on a 100 percent basis, except that the endurance portion of the performance test (see 3.10.7 for 5000 shot run) shall apply to one of each production lot, or to one of each fifty units, whichever is smaller.

4.3 Examination. The hypodermic injection apparatus shall be examined to determine that the materials, design, construction, physical dimensions, and workmanship are in accordance with applicable requirements. Visual and mechanical defects shall be classified during the course of the contract.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Leakage test, hydraulic valves, and seals. A completely assembled injection "gun", except for vaccine pump section, feed needle, and bottle holder, shall be adjusted so that the dosage setting is at the 1.0 cc. mark when the spring is fully cocked. The unit shall be left in the cocked condition for a period of 15 minutes. At the end of this period of time, there shall be no discernible creeping of the hydraulic piston as seen through the observation window. If creeping occurs, a faulty firing valve, check valve, quad or pin seal (part 80) is indicated. Manufacturer will be permitted to repair unit and resubmit it for retest.

4.4.2 Vaccine pump section test. The vaccine pump section shall be subjected to the following air pressure tests to determine leakage of valves, seals, or brazed joints.

4.4.2.1 Inlet valve, feed tube, brazed joints, seals (parts 93, 94, 96) test. An adapter, of aluminum alloy or other suitable metal, shall be fabricated by the contractor to permit air pressure to be applied to the inlet end of the vaccine pump cylinder (part 33), utilizing the coupling nut (part 34) to attach this adapter to the cylinder. A metal shim shall be placed in the injection nozzle blank (part 41) so that no air can leak out of the sapphire jewel orifice. If desired, a nozzle blank with no opening for the orifice may be used in this test. The vaccine pump assembly shall be filled with water, attached to air supply and then held vertically, nozzle down, immersed up to the coupling nut in water. An air pressure of 80 psi shall be applied to the assembly. There shall be no evidence of air leakage past the injection nozzle, feed tube joints, or inlet valve. A defective vaccine pump assembly may be repaired and submitted for retest.

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in 4.4.1.

Fahrenheit (F.) T.) F. when tested as specified

4.4.2.2 Outlet valve leak. An adapter of aluminum alloy, or other suitable metal, shall be fabricated by the contractor to permit air pressure to be applied in the reverse, or nonflow direction, to the outlet valve. This adapter shall be similar to the jet nozzle mount, except that the orifice shall be replaced by an air inlet fitting. The adapter shall screw onto the threads of the vaccine cylinder (part 33) and shall employ the teflon seal for sealing purposes. The outlet valve shall be filled with water prior to connecting to air source. The vaccine pump shall be immersed in water and an air pressure of 15 psi shall be applied to the outlet valve. There shall be no air leakage, as evidenced by air bubbles within the cylinder, as a result of this test. A defective outlet valve may be repaired and submitted for retest.

4.4.3 Feed needle test.

4.4.3.1 Leakage-air. An adapter, of aluminum alloy or other suitable metal, shall be fabricated by the contractor to permit air pressure to be applied to the inside of the hub (part 84) of the feed needle assembly. Air pressure shall be applied to check that the feed ports are open and that the feed needle annular passageway is open. Similarly, air pressure shall be applied to hub (part 83) to check that the air vent line is open. The feed needle shall be immersed in water, air pressure of 30 psi shall be applied to hub (part 84), and the outlet ports shall be sealed by closing them off by means of the test operator's fingers. There shall be no air leakage past the brazed joints of the feed needles as a result of this test. Similarly, the brazed joint in the air vent line shall be checked by applying air pressure at hub (part 83), and sealing off the air vent needle opening.

4.4.3.2 Restriction test. To insure that annular space between needles is free of foreign matter (flux, braze, or other foreign material), each needle shall be placed on an operating unit and used. The flow rate through the needle shall be such that the vaccine pump is not starved of solution when fired at the rate of 1200 shots per hour and the dosage adjustment is set at the 1-cc. setting.

4.4.4 Jet nozzle tests. The sapphire jewel orifice contours shall be checked by means of contour projection at 100 magnifications (using diiodomethane) to determine dimensional compliance with the detail drawing. After mounting of the sapphire, the jet nozzle shall be placed on an injection device and the injector fired a minimum of 25 times. The stream emitted shall be free from all side spray. There shall be no trace of fluid leakage around the periphery of the orifice insert or any loosening of the orifice insert in the mount. Each jet nozzle assembly including the spare shall be tested as described above.

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4.4.5 Hydraulic acceptance test. Random samples of the hydraulic power unit or power unit components shall be tested on a manufacturer's test stand for compliance with this specification. The contractor shall make available his or his supplier's test stand capable of performing the tests specified herein for this purpose. Units which fail may be reworked and resubmitted for retest. During this test, the hydraulic pump shall be capable of generating a minimum of 5000 psi with the hydraulic fluid temperature at 100° F. maximum. The pressure switch must be capable of operating at pressures from 500 to 5000 psi and shall be capable of being subjected to a pressure of 5000 psi without damage. Similarly, the relief valve shall be capable of operating in the 500 to 5000 psi range without damage. The completely assembled unit shall be capable of operating in ambient temperatures ranging from -30° F. to + 110° F. With an injection gun connected to the power unit, the gun shall be operated a minimum of 10 times to show that the pump is functioning properly and that relief valve and pressure switch are properly set. The unit shall be rotated about the motor axis and shall be turned end for end without any trace of hydraulic fluid leakage. The relief valve shall be set and checked to determine that it is at its specific setting prior to assembly of the power unit to the injection apparatus.

4.4.6 Performance test, hypodermic injection apparatus.

4.4.6.1 Leakage air test. The complete jet injection apparatus shall be set up to fire a bottle of water, following instructions given in the operator's technical manual. Water may be supplied from a continuous source. The injection apparatus shall be capable of sustained continuous performance. The injection apparatus shall be mounted in a test jig designed to fire the apparatus at the rate of 1200 shots per hour. Each unit manufactured shall be fired on a test stand for at least 750 shots. The unit under test shall fire at least 250 one shots and 500 1/2-cc. shots of water continuously without failure at the rate of 1200 shots per hour. In addition, one gun from each production lot shall be fired continuously for 5000 shots with the dosage set at 1-cc. There shall be no evidence of leakage or malfunctioning at the conclusion of the test. With dosage set at 1.00 cc., the vaccine pump shall purge itself of air and operate smoothly after 3 to 4 "shots". The vaccine stream shall be straight, without side streams, and shall cut off sharply at the end of the shot. There shall be no tailing which indicates the presence of air in the vaccine pump. All valves shall operate smoothly, and there shall be no hydraulic leakage past the gun stock valves, the observation window, the rear dosage adjustment knob, or the hydraulic hose connections to the gun. On cocking the gun, there shall be no drawback of fluid at the jet nozzle. After the gun is cocked, the solution bottle shall be removed, and the gun fired. There shall be no more than one drop of liquid drawn back through the vaccine pump inlet valve. The dosage adjustment device shall be checked for ease of adjustment and the unit shall be operated through several cycles at 0.2 and 0.5 cc. doses without failure. Note: One complete turn of the adjustment knob is equivalent to a 0.1 cc. change in dose. On cocking the unit, the pressure switch shall actuate the counter at the instant that the piston stops moving and just prior to the relief valve in the hydraulic power unit operating.

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4.4.6.2 Feed needle assembly test. In addition to the air test, each feed needle assembly (including spares) shall be subjected to a minimum of 25 shots without sign of any leakage, and with proper and sufficient flow of solution. The brazed points at either end of the inlet tube assembly (part 35) shall show no leakage at the conclusion of the performance test. Power for driving the gun can be supplied by the hydraulic pump itself or by the test stand. If the pump is used, disconnect the counter so that only a nominal amount of shots will be recorded on the counter prior to shipment. At the conclusion of testing, counter shall be reconnected and the gun fired at least five times prior to release of the unit.

4.4.7 Dielectric test. With all current-carrying parts of the apparatus in the circuit, a test potential of 900 volts shall be developed between the current-carrying parts and the non-current-carrying parts of the unit and maintained for one minute without electrical breakdown.

4.5 Government inspection. The contractor warrants that the contracting officer or his authorized representative shall, for the purpose of inspection, have access at all reasonable times to any plant (including the plant of any subcontractor) in which any of the work is being performed. In addition to inspection and testing required of the contractor, it is intended that in-process and final inspection will be conducted by the Naval Inspection Service under the technical direction of the Defense Personnel Support Center. Preproduction model inspection will be conducted by the Government.

5. PREPARATION FOR DELIVERY.

5.1 Packaging.

5.1.1 Level A. Component parts shall be arranged and cushioned within the carrying case in such a manner as to prevent shifting or damage. Carrying case shall be wrapped with waxed paper or other suitable material to prevent scratching or marring of finished surfaces.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Each wrapped unit shall be packed in a box of appropriate size constructed in accordance with PPP-B-636, compliance symbol V3c, style RSC. Box shall be provided with suitable exterior fittings, or shall contain sufficient cushioning material, to prevent shifting of, or damage to the contents.

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5.2.1.2 Closure. Closure shall be as specified in appendix, and waterproofing shall comply with 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Each wrapped unit shall be packed in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Box shall be provided with suitable interior fittings, or shall contain sufficient cushioning material, to prevent shifting of, or damage to, the contents.

5.2.2.2 Closure. Method I closure of Specification PPP-B-636 shall be utilized.

5.3 Marking.

5.3.1 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This hypodermic injection apparatus is intended for use in injecting immunizing agents rapidly, without using syringe and needle.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
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